AI/ML-Based Personalized Diet Plan Generator from Medical Reports

Milestone 4: Diet Plan Generation and UI Integration

Milestone Objectives

The objective of Milestone 4 was to integrate all previously developed components into a **fully functional end-to-end system** capable of generating personalized diet plans based on both structured and unstructured medical inputs.

As per the project roadmap, this milestone aimed to:

- Combine ML-based health analysis and AI/NLP-based text interpretation
- Generate personalized diet plans using integrated insights
- Implement a user-friendly frontend interface using React
- Enable export of diet plans in PDF and JSON formats
- Deliver a fully functional prototype tested with sample medical reports

System Integration Overview

By Milestone 4, the system combined three major components:

1. **Structured Data Analysis (Milestone 2)**
    - ML model (XGBoost) predicts diabetes risk
    - Provides probability score and risk level
    - Uses ROC-AUC as reliability metric
2. **Unstructured Text Analysis (Milestone 3)**
    - OCR extracts text from uploaded prescriptions
    - NLP interprets dietary instructions
    - Extracted rules mapped to diet guidelines
3. **Diet Plan Generation Engine (Milestone 4)**
    - Combines numeric risk analysis + extracted dietary rules
    - Generates structured daily meal plans
    - Displays results in a modern frontend UI

This milestone completes the full pipeline from **input → analysis → interpretation → personalized output**.

Personalized Diet Plan Generation Logic

The diet generation module uses:

- ML-based diabetes risk level
- Extracted prescription instructions
- Predefined nutritional rules
- Balanced macronutrient distribution

Based on the combined insights, the system generates a complete daily meal plan structured into:

- **Breakfast**
- **Lunch**
- **Dinner**
- **Recommended Snacks**

Each meal includes:

- Food item name
- Description
- Calorie value
- Macronutrient indicators (carbohydrates, protein, fat)

The plan adapts based on:

- Risk level (Low / Moderate / High)
- Dietary restrictions (e.g., avoid sugar, low carb, high fiber)
- General health recommendations

Frontend Implementation (React UI)

The frontend was implemented using React to provide an interactive and user-friendly interface.

Core UI Features:

- **Input form for health parameters**
- **Prescription image upload**
- **Real-time risk prediction display**
- **Dynamic meal plan rendering**
- **Color-coded risk visualization**

- **Export options**

The UI ensures that users can easily:

- **Enter medical data**
- **Upload prescriptions**
- **View personalized recommendations**
- **Download results**

<u>Export Functionality</u>

To improve usability and portability, the system supports:

1. PDF Export

- Allows users to download their personalized meal plan
- Useful for offline access or medical consultation

2. JSON Export

- Structured output format
- Enables future integration with:
  - Mobile apps
  - Nutrition tracking tools
  - Healthcare systems

<u>End-to-End Prototype Validation</u>

The system was tested using:

- Sample numeric medical inputs
- Sample prescription images
- Extracted dietary instructions

Validation Results:

- Risk prediction successfully generated
- OCR extracted readable text
- NLP mapped dietary instructions correctly
- Personalized meal plan generated dynamically
- Export functionality worked successfully

The system achieved a **fully functional end-to-end prototype** meeting the milestone criteria.

System Architecture Completion

By the end of Milestone 4, the following modules are fully integrated:

- Medical Data Input Module
- OCR Extraction Module
- NLP Interpretation Engine
- ML-Based Health Risk Analysis
- Personalized Diet Generation Engine
- React-Based Frontend Interface
- Export (PDF & JSON) Module

This completes the core architecture of the AI Diet Planner system.

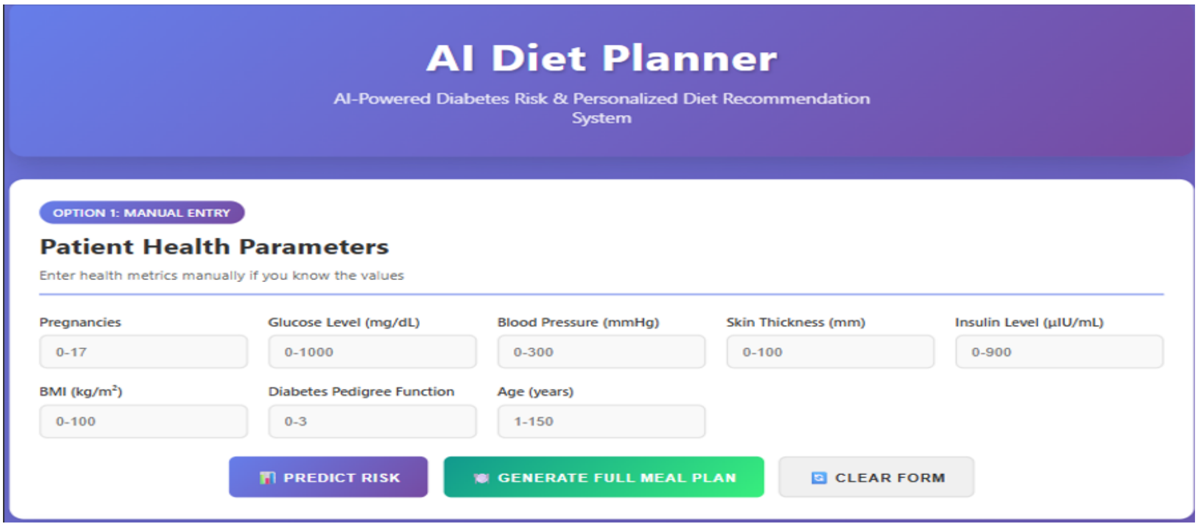
UI Demonstration

Figure: End-to-End AI Diet Planner Interface with Generated Meal Plan

The screenshot demonstrates:

- Extracted medical information
- Generated breakfast, lunch, dinner, and snack recommendations
- Calorie and nutrient indicators
- Export options

OPTION 1: Manual Entry



OPTION 1: MANUAL ENTRY

Patient Health Parameters

Enter health metrics manually if you know the values

Pregnancies	Glucose Level (mg/dL)	Blood Pressure (mmHg)	Skin Thickness (mm)	Insulin Level (µIU/mL)
1	400	100	30	400

BMI (kg/m²)	Diabetes Pedigree Function	Age (years)
40	1	45

PREDICT RISK | GENERATE FULL MEAL PLAN | CLEAR FORM

Prediction Results

RISK LEVEL:	DIABETES PROBABILITY:	MODEL ROC-AUC SCORE:
High Risk	78.7%	0.81

RECOMMENDATION:

Follow a low-GI diet and consult a doctor

Personalized Meal Plan

Based on your health parameters and diabetes risk

Risk Level: High Risk

Diabetes Probability: 78.7%

MONDAY

BREAKFAST	Oatmeal with berries and nuts	350 cal
LUNCH	Grilled chicken salad	420 cal
SNACK	Whole grain crackers with cheese	140 cal
DINNER	Baked salmon with steamed vegetables	450 cal
DAILY TOTAL	~1360 cal	

TUESDAY

BREAKFAST	Vegetable omelet with whole wheat toast	320 cal
LUNCH	Quinoa bowl with roasted vegetables	380 cal
SNACK	Greek yogurt	120 cal
DINNER	Grilled chicken with brown rice	480 cal
DAILY TOTAL	~1300 cal	

WEDNESDAY

BREAKFAST	Greek yogurt with chia seeds	280 cal
LUNCH	Lentil soup with vegetables	350 cal
SNACK	Fresh fruit	80 cal
DINNER	Turkey meatballs with zucchini noodles	420 cal
DAILY TOTAL	~1130 cal	

THURSDAY

BREAKFAST	Oatmeal with berries and nuts	350 cal
LUNCH	Grilled chicken salad	420 cal
SNACK	Handful of nuts	160 cal
DINNER	Baked salmon with steamed vegetables	450 cal
DAILY TOTAL	~1380 cal	

FRIDAY

BREAKFAST	Vegetable omelet with whole wheat toast	320 cal
LUNCH	Quinoa bowl with roasted vegetables	380 cal
SNACK	Vegetable sticks with hummus	100 cal
DINNER	Grilled chicken with brown rice	480 cal
DAILY TOTAL	~1280 cal	

SATURDAY

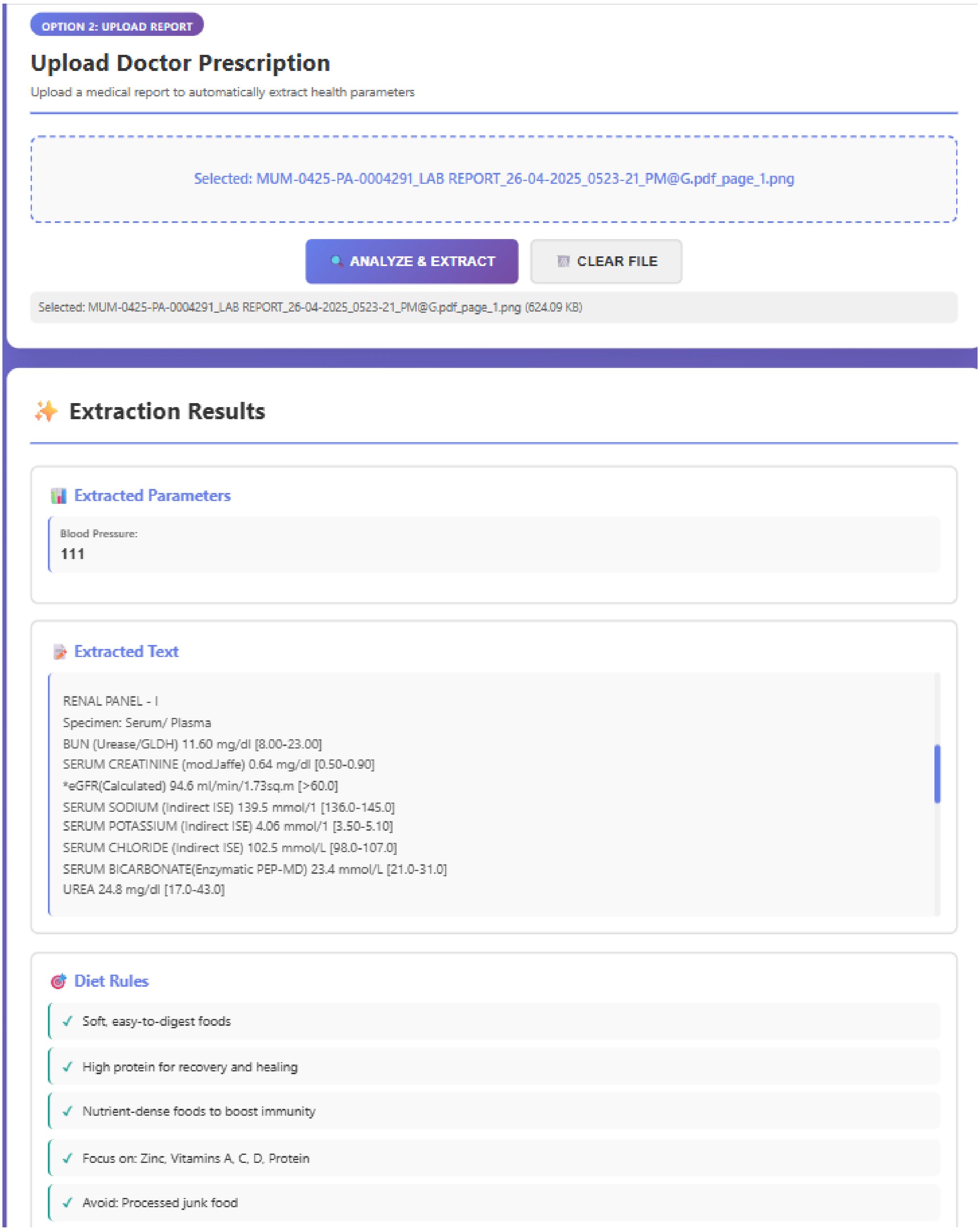
BREAKFAST	Greek yogurt with chia seeds	280 cal
LUNCH	Lentil soup with vegetables	350 cal
SNACK	Whole grain crackers with cheese	140 cal
DINNER	Turkey meatballs with zucchini noodles	420 cal
DAILY TOTAL	~1190 cal	

SUNDAY

BREAKFAST	Oatmeal with berries and nuts	350 cal
LUNCH	Grilled chicken salad	420 cal
SNACK	Whole grain crackers with cheese	140 cal
DINNER	Baked salmon with steamed vegetables	450 cal
DAILY TOTAL	~1360 cal	

EXPORT AS PDF | EXPORT AS JSON

OPTION 2: Upload Doctor Prescription



🍽️ 7-Day Personalized Meal Plan

Complete weekly meal plan with breakfast, lunch, snack, and dinner

MONDAY

BREAKFAST
Oatmeal with berries & almonds
350 cal · 12g protein

LUNCH
Grilled chicken breast with brown rice
520 cal · 45g protein

SNACK
Apple with almonds
180 cal · 6g protein

DINNER
Baked lean fish with sweet potato
550 cal · 48g protein

Daily Total: 1600 cal

TUESDAY

BREAKFAST
Scrambled eggs with whole wheat toast
300 cal · 18g protein

LUNCH
Salmon fillet with steamed broccoli
480 cal · 42g protein

SNACK
String cheese & crackers
150 cal · 8g protein

DINNER
Grilled chicken with quinoa
580 cal · 50g protein

Daily Total: 1510 cal

WEDNESDAY

BREAKFAST
Greek yogurt with granola
280 cal · 15g protein

LUNCH
Turkey sandwich with lettuce
420 cal · 35g protein

SNACK
Carrot sticks with hummus
120 cal · 4g protein

DINNER
Beef stir-fry with vegetables
520 cal · 46g protein

Daily Total: 1340 cal

THURSDAY

BREAKFAST
Smoothie bowl with banana & chia
320 cal · 10g protein

LUNCH
Chickpea Buddha bowl
480 cal · 18g protein

SNACK
Mixed nuts
170 cal · 6g protein

DINNER
Vegetable curry with lentils
480 cal · 20g protein

Daily Total: 1450 cal

FRIDAY

BREAKFAST
Whole wheat pancakes with honey
340 cal · 14g protein

LUNCH
Tuna salad with olive oil
450 cal · 38g protein

SNACK
Protein bar
200 cal · 10g protein

DINNER
Turkey meatballs with pasta
520 cal · 42g protein

Daily Total: 1510 cal

SATURDAY

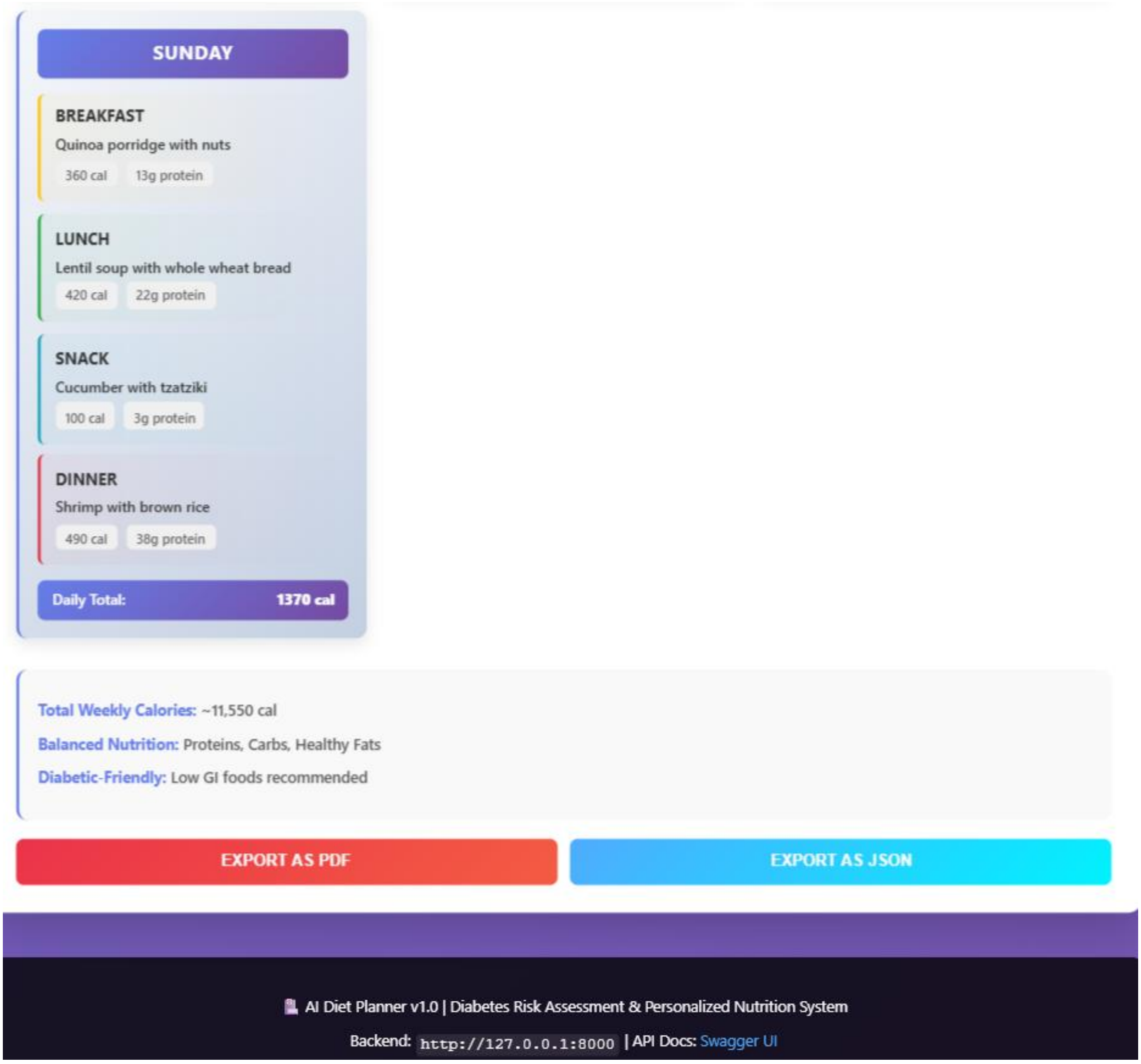
BREAKFAST
Vegetable omelette with mushrooms
280 cal · 16g protein

LUNCH
Tofu stir-fry with vegetables
380 cal · 28g protein

SNACK
Berries with yogurt
140 cal · 7g protein

DINNER
Baked chicken breast with veggies
500 cal · 52g protein

Daily Total: 1300 cal



Key Learnings

Through Milestone 4, the following skills and concepts were strengthened:

- Full-stack AI application development
- Integration of ML + NLP pipelines
- Backend-frontend API communication
- User-centered UI design
- Deployment-ready architecture design
- End-to-end healthcare system prototyping

Conclusion

Milestone 4 successfully integrated machine learning-based health analysis and NLP-based prescription interpretation into a complete, functional system capable of generating personalized diet plans. The frontend implementation provides an intuitive interface, while the backend ensures intelligent processing and reliable results.

The system now operates as a **fully functional end-to-end prototype**, tested with sample medical data and prescription reports. This milestone marks the completion of the core implementation phase of the AI Diet Planner project.

SUBMITTED BY:

Sabiha Anjum